

Writing Macros I

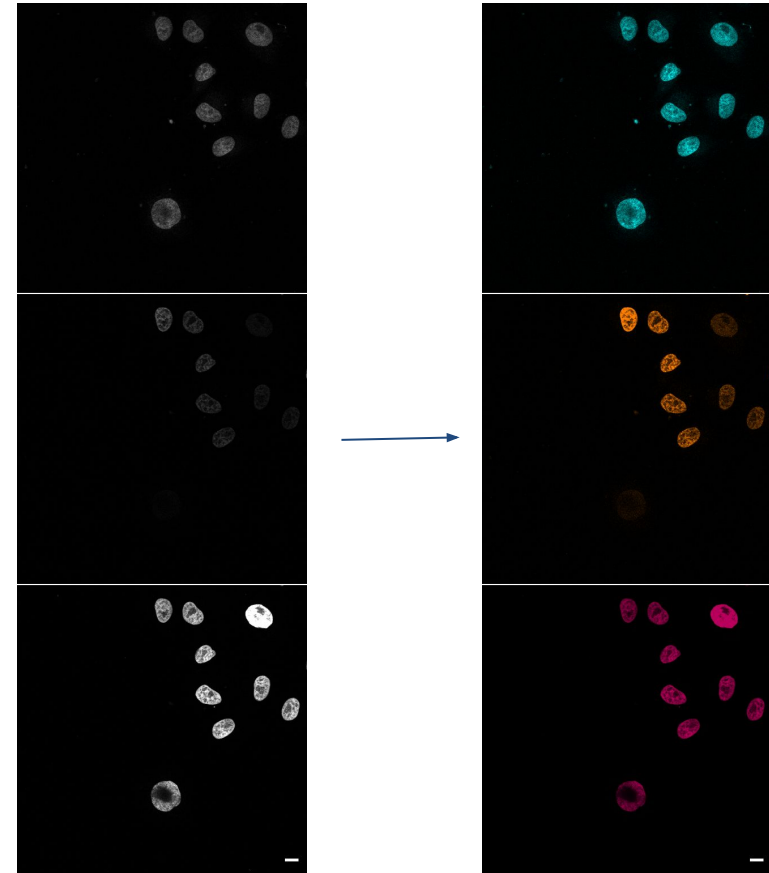
Image Processing & Analysis for Life Scientists

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Writing your own Macro

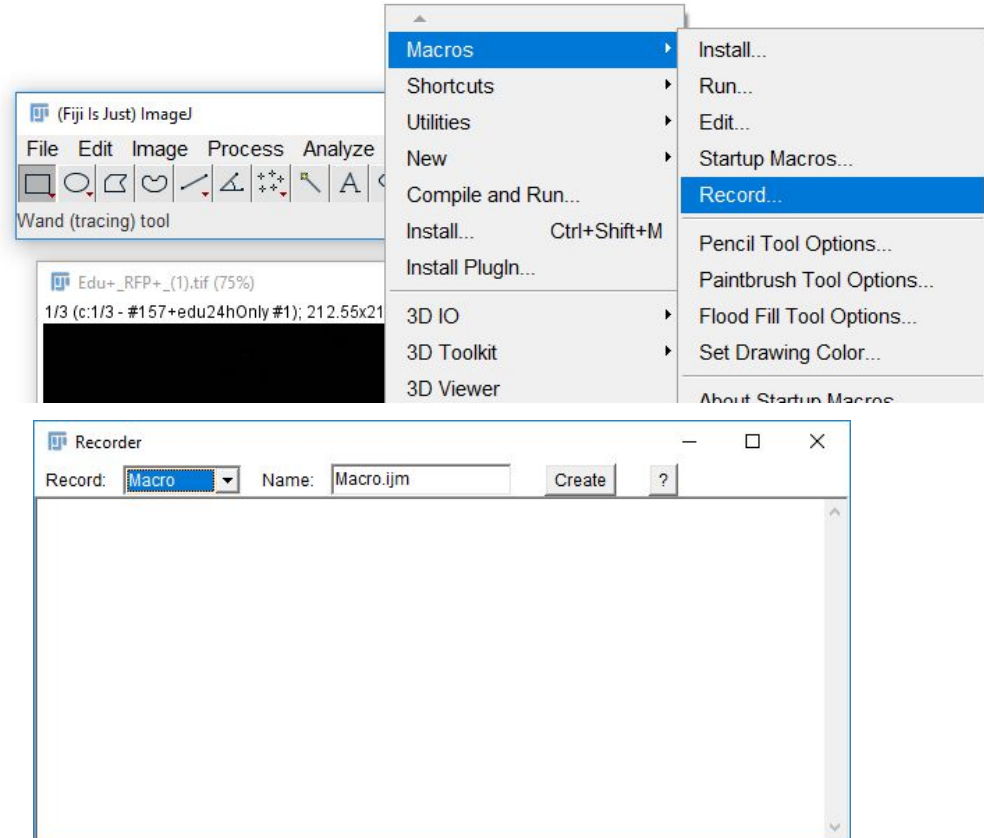
Suppose we have a series of images on which we want to:

1. Apply specific lookup tables
2. Adjust the brightness & contrast for each channel
3. Show all channels at once



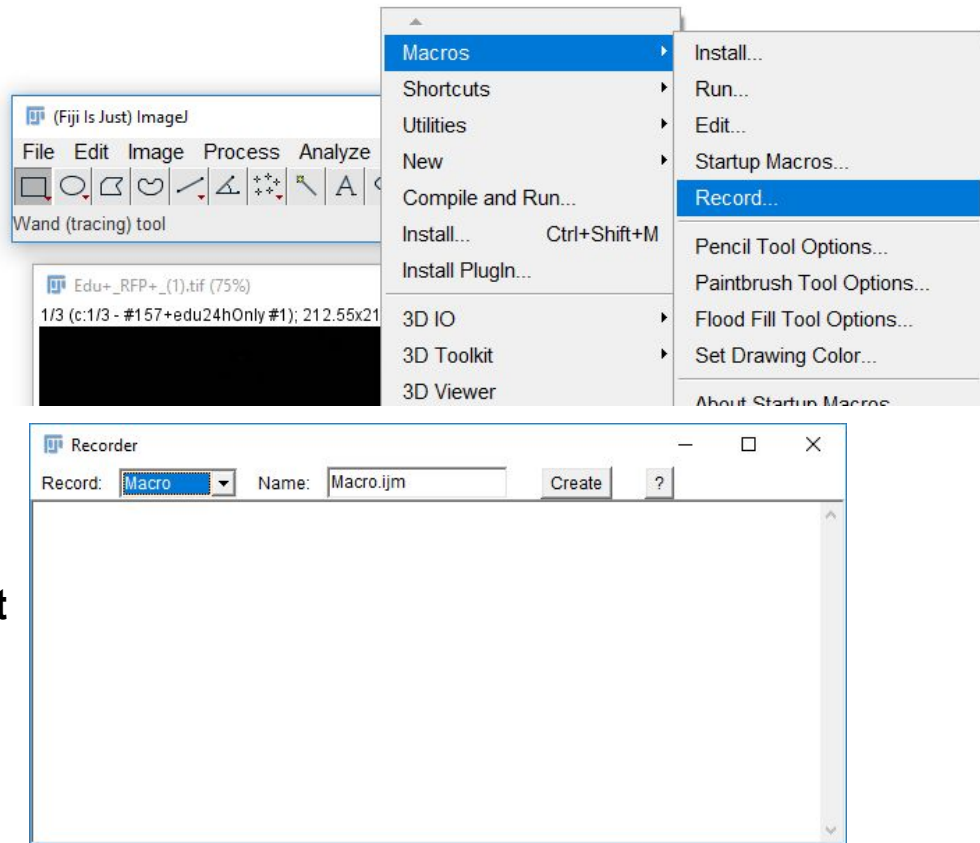
Writing Macros

- Download and open an image from [this dataset](#)
- Go to Plugins>Macros>Record...

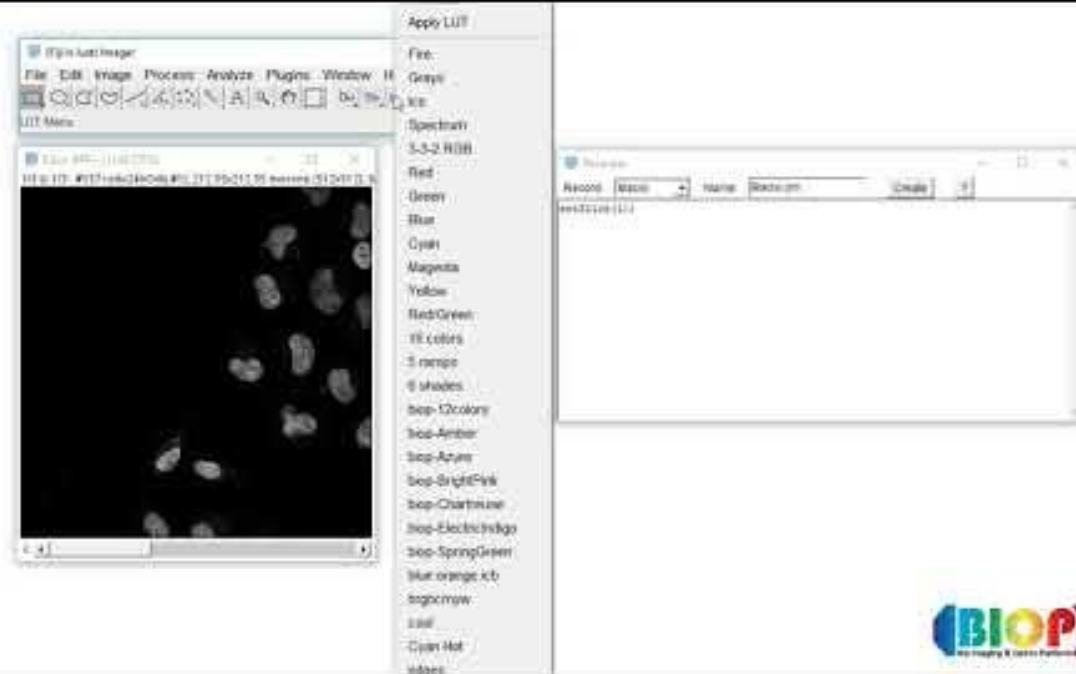


Writing Macros

- Download and open an image from [this dataset](#)
 - Go to **Plugins>Macros>Record...**
 - **Set the Brightness & Contrast + LUT**
 - Go to **Image>Stacks>Set Slice...**
And choose slice 1
 - Go to **Image >Lookup Tables**
And choose **Cyan**
 - Go to **Image>Adjust>Brightness & Contrast**
Adjust the min and max sliders
Then press “Set” and OK
- See video on next slide for help



Setting LUT & Brightness/Contrast Example



- Set the image to “Composite”
Image>Color>Make Composite
- Your Recorder should look like this



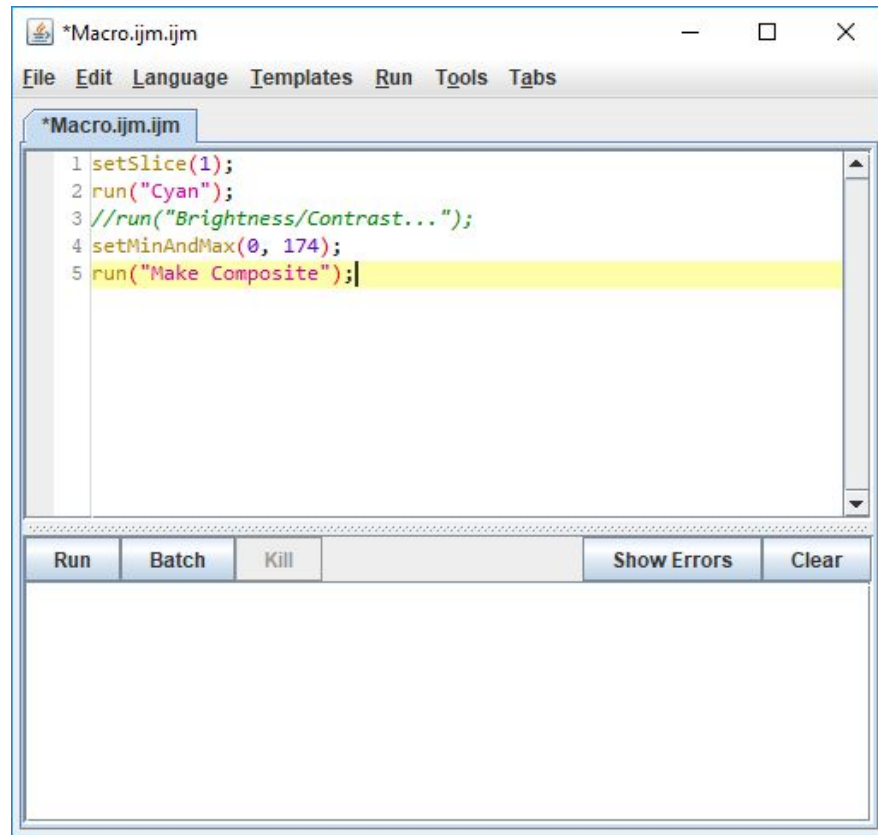
Writing Macros

- Set the image to “Composite”
Image>Color>Make Composite
- Your Recorder should look like this
- Click on “Create” to start writing your macro from this template

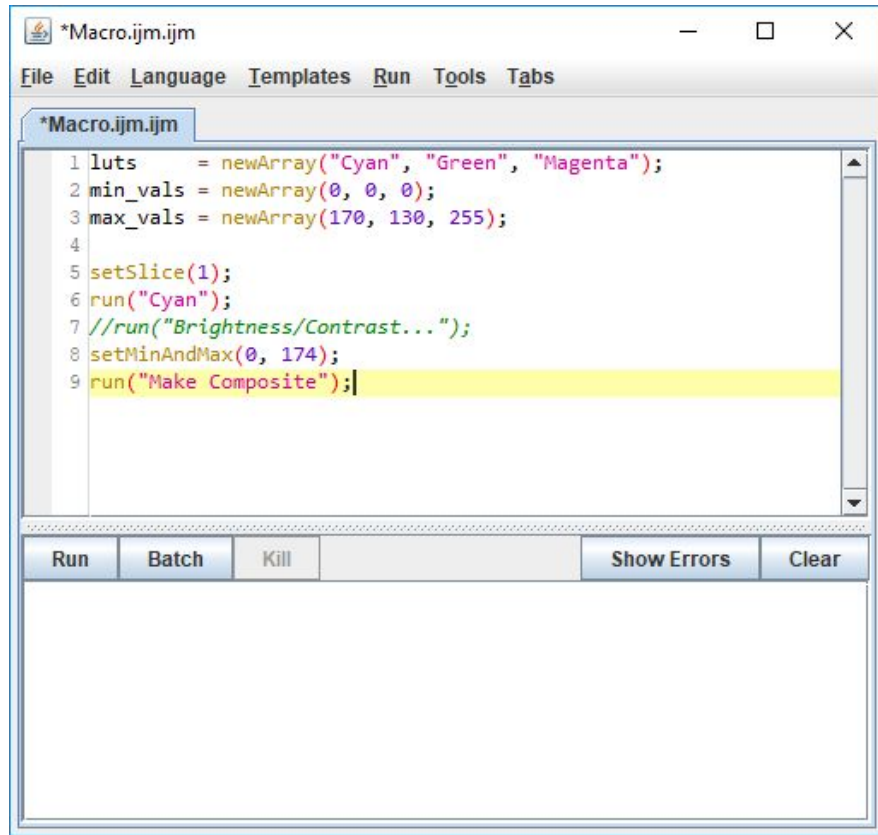


Editing Macros

- Create 3 arrays
 - a. to store the Lookup tables for each channel
 - b. To store the min value for each channel
 - c. To store the max value for each channel
- Select values for each element of the array



- Create 3 arrays
 - a. to store the Lookup tables for each channel
 - b. To store the min value for each channel
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- Select values for each element of the array



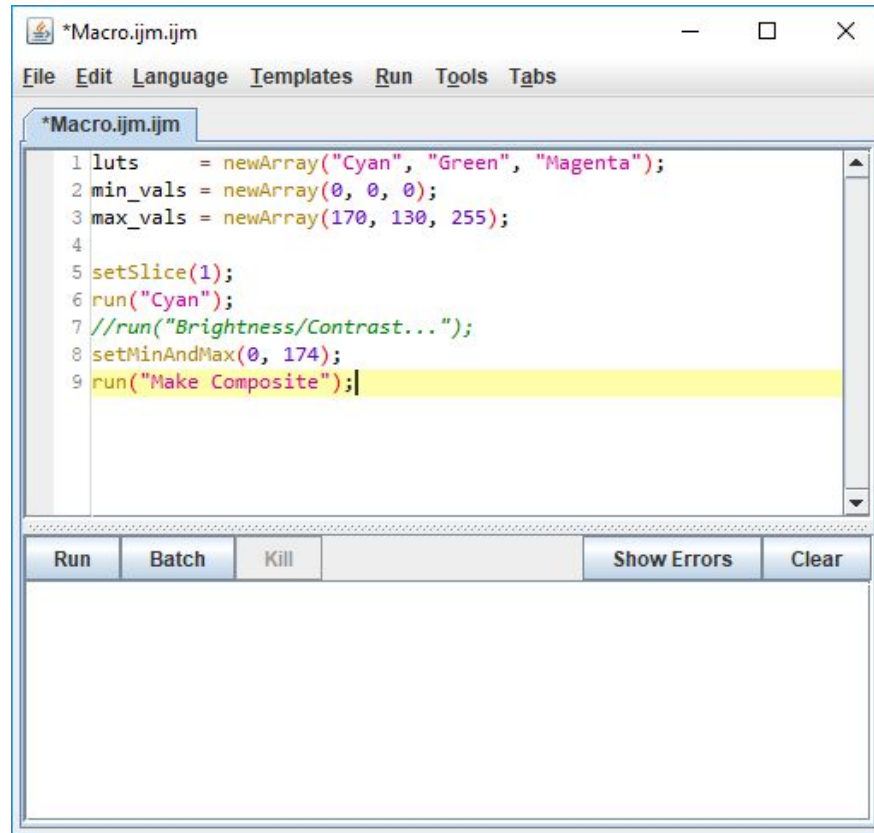
```
*Macro.ijm.ijm
File Edit Language Templates Run Tools Tabs

*Macro.ijm.ijm
1 luts      = newArray("Cyan", "Green", "Magenta");
2 min_vals = newArray(0, 0, 0);
3 max_vals = newArray(170, 130, 255);
4
5 setSlice(1);
6 run("Cyan");
7 //run("Brightness/Contrast...");
8 setMinAndMax(0, 174);
9 run("Make Composite");|
```

Run Batch Kill Show Errors Clear

Editing Macros

- Create a loop to go through the channels. The loop will
 - a. Set the slice
 - b. Apply the LUT
 - c. Set the min and max
- Place the part that does the work in a loop



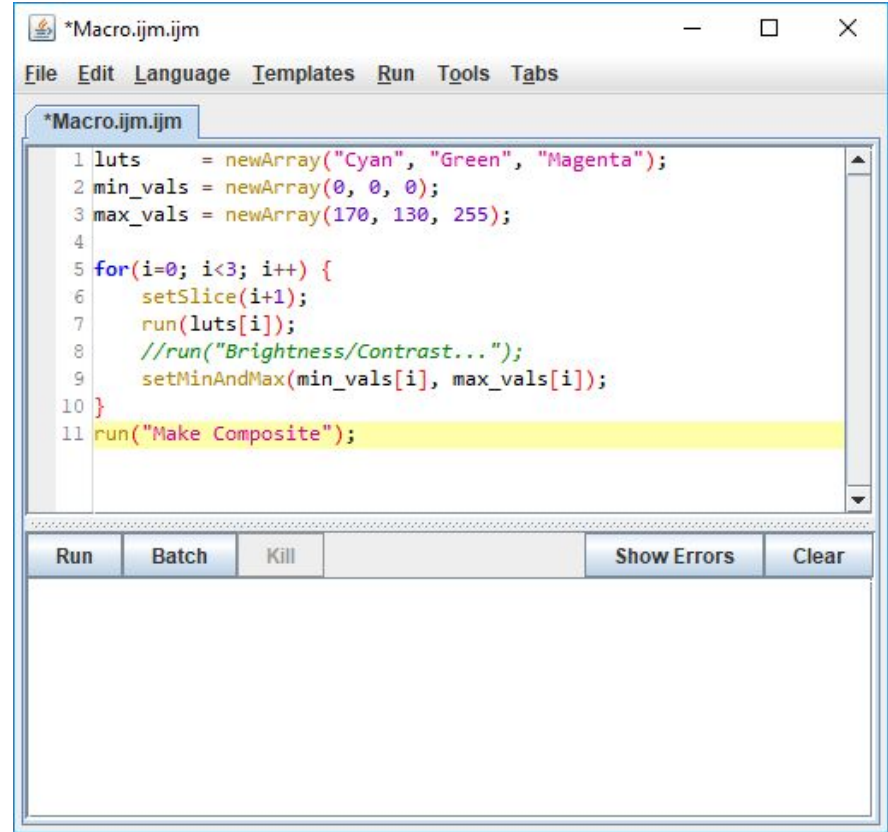
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Run Batch Kill Show Errors Clear

Editing Macros

- Create a loop to go through the channels. The loop will
 - a. Set the slice
 - b. Apply the LUT
 - c. Set the min and max
- Place the part that does the work in a **for** loop
- Replace the values of **setSlice**, **run** and **setMinAndMax** to the array indexes



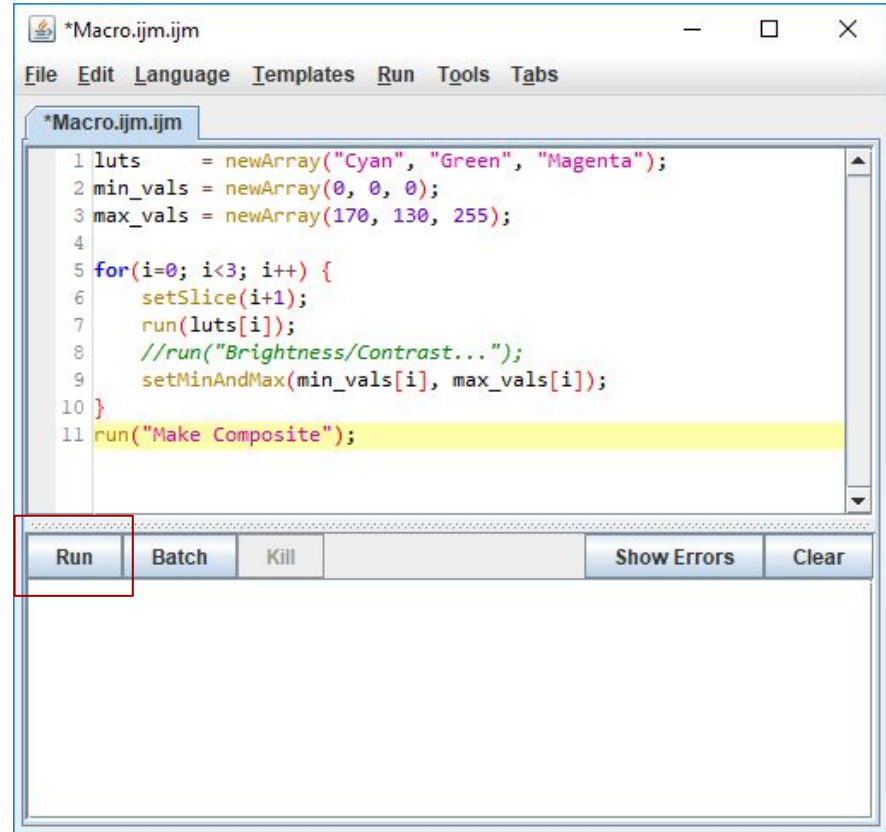
```
*Macro.ijm.ijm
File Edit Language Templates Run Tools Tabs

*Macro.ijm.ijm
1 luts      = newArray("Cyan", "Green", "Magenta");
2 min_vals = newArray(0, 0, 0);
3 max_vals = newArray(170, 130, 255);
4
5 for(i=0; i<3; i++) {
6     setSlice(i+1);
7     run(luts[i]);
8     //run("Brightness/Contrast...");
9     setMinAndMax(min_vals[i], max_vals[i]);
10 }
11 run("Make Composite");
```

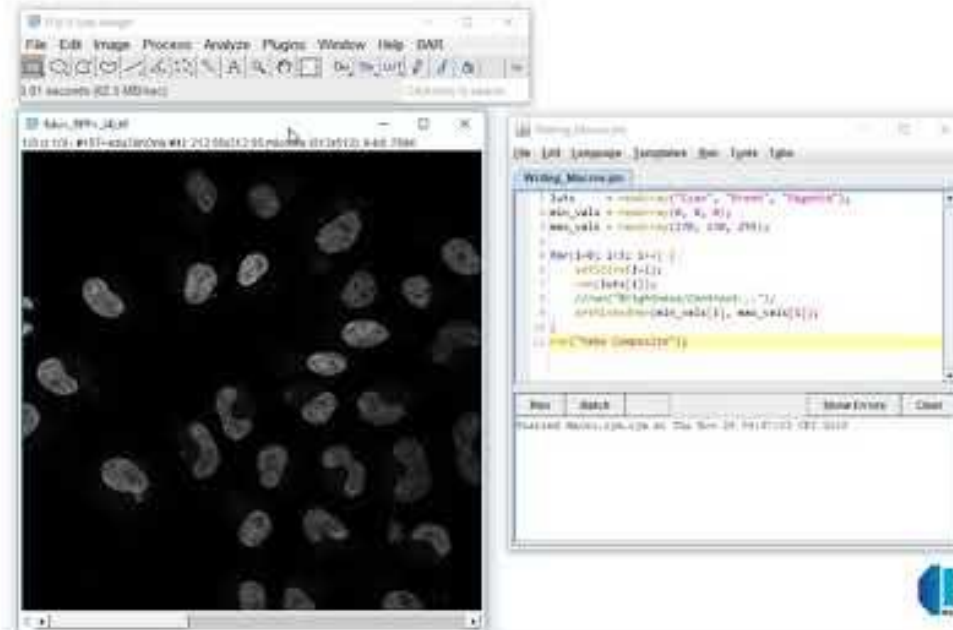
Run Batch Kill Show Errors Clear

Save and Apply

- Save your macro with **File>Save as**
- Open another image
- Hit **Run** on the Script Editor



Running the Macro



Save and Apply
